

UK Train Rides Data Analysis

A data analysis project exploring UK train ride patterns, delays, and performance using Python, SQL, and Power BI."

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GitHub Repository:

UK_Train_Rides

https://github.com/MaihOub/UK_Train_Rides

Abstract

This project focuses on analyzing UK train rides to understand travel patterns, delays, performance issues, seasonal trends, and passenger behavior.

Using Python, SQL, and Power BI, the team performed a full data analysis workflow starting from dataset collection, cleaning, validation, exploratory analysis, and visualization to generate meaningful insights that support decision-making and improve train service performance.

Introduction

Train transportation represents a major part of the UK mobility system.

Understanding trends in delays, ticket usage, peak times, and demand helps companies improve performance and reduce passenger dissatisfaction.

This project aims to analyze real-world UK train-related data using Data Analysis techniques and powerful tools such as Python, SQL, and Power BI.

Problem Statement

The UK rail system frequently experiences:

- Delays
- High passenger demand at peak hours
- Variations in ticket types usage
- Seasonal fluctuation in rides

A structured analysis is required to understand the factors behind these patterns.

Project Objectives

- Collect and clean UK train rides dataset

- Validate the dataset to ensure accuracy and consistency
 - Perform Exploratory Data Analysis (EDA)
 - Analyze peak hours, delays, ticket usage, and demand patterns
 - Answer business questions through insights
 - Visualize findings using Python and Power BI
 - Support future improvements in scheduling and capacity planning
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Tools and Technologies Used

Languages:

- Python (Pandas, Matplotlib)
- SQL

Platforms:

- Jupyter Notebook
 - Power BI
 - GitHub
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Dataset Description

The dataset includes details such as:

- Train ride dates
- Start and end stations
- Duration
- Ticket types
- Delay times
- Passenger counts
- Travel class
- Price information

Dataset Source: Collected online from public UK transport datasets (or mention your actual source).

Data Collection

The dataset was gathered from kaggle

Each team member contributed to verifying the data structure and format (CSV / XLSX).

Data Cleaning

The team performed several cleaning steps:

- ✓ Handling missing values
- ✓ Removing duplicates
- ✓ Converting data types (dates, integers, categories)
- ✓ Standardizing station names
- ✓ Fixing formatting issues
- ✓ Removing invalid entries (negative delays, impossible durations)

All cleaning steps were done using **Python (Pandas)**.

Data Validation

SQL was used to validate the cleaned dataset through:

- Checking NULL ratio
- Verifying primary keys
- Ensuring consistent formatting
- Running summary statistics
- Identifying outliers

Examples:

- Ensure delay_time ≥ 0
- Ensure ticket_type IN ('Standard', 'Peak', 'Off-Peak')
- Ensure start_station $\neq$ end_station

Exploratory Data Analysis (EDA)

The team analyzed several aspects:

✓ Delay patterns

- Average delay per station
- Delays by time of day
- Delays by day of week

✓ Passenger demand

- Busiest routes
- Peak hours
- Monthly & seasonal variations

✓ Ticket analysis

- Ticket types usage
- Price comparisons
- Class distribution

✓ Correlations

- Relationship between distance, duration, and price
- Delay vs passenger count

EDA was performed using Python visualizations.

SQL Analysis

SQL queries were created to answer business questions:

- Most crowded routes
- Total number of monthly rides
- Stations with highest delays
- Top revenue-generating routes
- Average ticket price per class

The SQL file is included in the GitHub repo.

Python Analysis

A Python notebook was developed that includes:

- Data loading
 - Cleaning functions
 - EDA plots
 - Statistical summaries
 - Feature extraction (peak-hour flag, delay category)
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Visualization

Python Visualization

Using Matplotlib/Seaborn:

- Bar charts for crowded stations
- Line charts for monthly trends
- Heatmaps for correlation
- Pie charts for ticket distribution

Power BI Dashboard

Interactive dashboard showing:

- Number of rides
 - Delay analysis
 - Route performance
 - Ticket sales
 - Seasonal demand trends
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Insights (Important for the report)

Top Insights Found:

- Peak hours are usually between **7-9 AM** and **5-7 PM**
- Some stations consistently show higher delays
- Off-Peak tickets are more common during weekends
- Higher passenger count correlates with longer delays
- Summer months show increased travel activity

- Certain routes have significantly higher revenue

(You can edit these based on your actual results.)

Conclusion

The project successfully analyzed UK train ride data and extracted valuable insights that can help improve train scheduling, reduce delays, and enhance customer satisfaction.

Future improvements can include real-time data integration and predictive analytics.

Future Work

- Build a predictive model to forecast delays
- Improve data quality with real-time APIs
- Add weather data to analyze its effect on delays
- Expand the dashboard for more KPIs

References

- UK Transport Open Data Sources
- Python Documentation
- Power BI Resources
- GitHub repository files
- Team proposal document